LECTURE / SESSION[1] PLAN

Name Lecturer: Sofia Gil-Clavel	**Date session:** 10	**Expected number of students:** 20
Course title	R-Workshop	
Topic of lecture or session	Basic Programming	
Situational factors (e.g. group size, prior knowledge, expected motivation)	The class lasts 3hrs. The students are in different career stages, from PhD students to professors. The students have followed the previous sessions of the course or have experience with the topics. The students use different Operating Systems (Windows, Mac, or Linux).	
Intended learning outcomes of this session	At the end of the session the students will be able to: 1. Situate all what they have learned during the course in the context of digital data analysis. 2. Explain and use APIs to retrieve data. 3. Write functions 4. Write branches and loops in R. 5. Perform error handling. 6. Put everything together to continuously retrieve data using APIs.	
Learning material (book, chapters, ...)	The lecture is based on the book: • Davies, Tilman M. ***The Book of R: A First Course in Programming and Statistics***. No Starch Press, 2016.	
Media, equipment, tools	The students use their own laptops. The teacher needs access to a projector and a whiteboard.	

[1] A session means a teaching and learning session for university students in the bachelor or master. It can be a lecture, a seminar or a specific type of educational meeting with a group of students that is relevant for the discipline where the lecturer can demonstrate a whole range of teaching skills. It means that in one session each of the six didactic elements appears at least once. Your lesson plan should show that you are using a powerful learning environment and that students are activated.

Preparation for students	The students have access to the slides and codes before the class: https://github.com/SofiaG1l/R_Course/tree/master/10_Session_BasicProgramming The students need to apply and obtain a The Guardian developer key here: https://open-platform.theguardian.com/access/

Time[2] **(min.)**	**Didactic element (goal)**[3] **and topic**[4]	**What the teacher does**[5] (teacher activity)	**What students do** (learner activity)	**Evaluation**[6] (feedback/assessment)
20	Situate all what they have learned during the course in the context of digital data analysis.	• The teacher uses the slides to remind the students what they have learned during the course. • The teacher explains how all theses topics tie together to do analyse digital data.	• The students passively digest what the teacher is explaining.	• The teacher gives students the opportunity to ask any questions.
30	Explain and use APIs to retrieve data.	• The teacher uses the slides to explain what an API is and how it works. • The teacher uses the slides to explain how to interact with APIs using the package httr2.	• The students passively digest what the teacher is explaining. • The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides. • The students retrieve their first The Guardian data.	• • The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.

[2] Indicate the planned duration in minutes
[3] State the number(s) of the relevant ILO at each didactic element
[4] Only key words
[5] Specify the type of activity and write down also the questions that you prepared to ask in the session
[6] Specify the type of evaluation, i.e. the way in which you assess if the objective(s) has/have been achieved

30	Write functions	• The teacher uses the slides to explain how to write functions.	• The students passively digest what the teacher is explaining. • The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides. • The students use their new knowledge to turn their The Guardian retrieval code into a function.	• The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.
30	Explain what conditions and loops are.	• The teacher uses the slides to motivate and explain what branches and loops are. • The teacher opens RStudio and opens the already written script that will be used during the class.	• The students passively digest what the teacher is explaining. • The students open and follow the steps that the teacher is explaining.	• The students will use these concepts during the workshop. So, the teacher will detect when a student confuses them. Based on these confusions the teacher will be able to correct the student.

30	Perform error handling.	• The teacher uses the slides to explain how the possible sources of errors when using APIs to retrieve data.
		• The teacher explains how these errors can be handled using:
		    ○ try()
		    ○ Sys.sleep()
		Together with conditions and loops.
			• The students passively digest what the teacher is explaining.
			• The students independently start filling out the missing parts of the R-script. This is based on what the teacher explained using the slides.
				• The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.

30	Put everything together to continuously retrieve data using APIs	• The teacher gives students time to think how to assemble a loop to continuously retrieve data based on what they learned during the session.	• The students independently start working on their R-script.	• • The teacher walks around the classroom to check on the students and provide feedback when something is not working on their computers. When the teacher detects a common error, then the teacher uses the whiteboard to clarify.